PHILOSOPHY 1002: INTRODUCTION TO CRITICAL THINKING
Workshops for Part 2

In most of these weeks there are multiple sets of questions. If you do not complete them all, do the others as assignment questions for that week.

Week 7: Definitions & Clarity

(Exercise a) Explain the difference between a reportive and a stipulative definition. Can either go wrong – and if so, how? Can you think of a novel stipulative definition that could be useful?

(Exercise b) (i) Explain, using examples, the difference between an essential and a criterial definition. What is an operational definition? (ii) Must a good reportive definition also be an essential definition?

(Exercise c) Govier, Ch. 3, p79, Q2: g, with the modified instruction: *Assume that the following statement is put forward as offering reportive definition. Identify any problems.*

(Exercise d) Govier, Ch. 3, p80, Q3: g, l & m.

(Exercise e) Read the following passage and assess whether it contains any examples of *ambiguity* or *vagueness*. Explain any words that you think may give rise to the ambiguity and try to state the possible different meanings and see if you can reconstruct the argument so as to reveal a related fallacy. If you think it contains vagueness, explain where the vagueness arises. Identify any other fallacies of clarity.
This is an ad that appeared in *Harper's* magazine in October 1978:

"Mother Nature is lucky her products don't need labels. All foods, even natural ones, are made up of chemicals. But natural foods don't have to list their ingredients. So it's often assumed that they're chemical-free. In fact, the ordinary orange is a miniature chemical factory. And the good old potato contains arsenic among its more than 150 ingredients. This doesn't mean natural foods are dangerous. If they were, they wouldn't be on the market. All man-made foods are tested for safety. And they often provide more nutrients, at a lower cost, than natural foods. They even use many of the same chemical ingredients. So you see, there really isn't much difference between foods made by Mother Nature and those made by man. What's artificial is the line drawn between them."

(Exercise f) Discuss this example. Try to locate in it a problem with definitions, and to see how the reasoning is affected by that problem:
"We find it self-evident that women are a colonised group who have never – anywhere – been allowed self-determination. Therefore, all women who fight against their own oppression (colonised status) as females under male domination are anti-imperialist by definition." [Barbara Burris, *The Fourth World Manifesto*]

Week 8: Fallacies of Clarity & Relevance

(Exercise a) In this passage, determine the conclusions, if any or if several, and then assess the reasoning for relevance, checking for fallacies (if any):

1. [Background: In an essay published in Harper's Magazine, Michael Manley, a former Prime Minister of Jamaica, wrote that poverty in the Third World was a result of imperialism in the colonising of many countries during the nineteenth century. Thomas Sowell, of the Hoover Institution, replied, objecting to Manley's theory.] Sowell wrote:

"The notion that either the capital or the standard of living in the West depends upon the Third World will not stand the slightest contact with evidence. When the British Empire was at its peak, around the

First World War, more British Capital was invested in the United States than in all of Africa or all of Asia outside of Australia. The French had more trade with little Belgium than all its farflung African empire. Germany's trade with its colonies was less than 1% of its exports."

(Exercise b) Fill in the missing disjunctive premise in these false dichotomies, and then explain why it is a false dichotomy using a counterexample. Is the missing premise a contrary statement?

1. [Background: Janet and Clare are about to go out on the town. Clare asks Janet if she needs to take a raincoat.]

Janet: "It's not stormy weather outside right now. So it is clear skies!"

2. [Background: Over the summer months, Scarborough holds outdoor markets every Thursday evening near its foreshore with an atmosphere that is quintessentially 'holigetic' (meaning a buzzing vibrant holiday energy). Bob has just flown in and has lost track of what day it is.]

Bob: "There's a holigetic vibe around Scarborough tonight, so it must be a Thursday evening."

(Exercise c) Govier, Ch. 6 pp. 167–70; Exercise 2: Q. 2, Q.5, Q8, Q12.
[For Q.12 Check the answer in the book (see p. 395). What do you think?]

(Exercise d) Govier, Ch. 6, pp. 152–54; Exercise 1: part B: Q. 10

Extras Exercises (to supplement exercise a):
In this passage, determine the conclusions, if any or if several, and then assess the reasoning for relevance, checking for fallacies (if any):

"[An article in the World press Review for May 1994 entitled] 'Pollution: The Price of Progress' states that since Europe and the United States polluted heavily during the Industrial Revolution, Asia and the developing world have a right to do the same. Developing nations will serve themselves – and the world – best by learning from the mistakes of the industrial world. He who does not learn from the past is condemned to repeat it, and it is damnation indeed to repeat the ecological devastations of the past."

Week 9: Test #1

Before your workshop group meets be sure you have (i) done the take-home test under the specified examination conditions and submitted it by the date shown on the Unit Diary; and (ii) evaluated your own performance by checking your work against the answer key posted on LMS.
Bring your questions about the test and your performance to the workshop for discussion.
Note: The workshop will not consist of your tutor simply going over the entire test. It will consist of dealing with specific queries students have about the test.

Week 10: Analogies

Here are some terms and definitions you'll need for this week's workshop. These can all be found in the Govier text, but I've collected them here so that you can easily access them.
An **argument by analogy** is a form of argument that begins by using one case (usually agreed on and relatively easy to understand) to illuminate or clarify another (usually less clear). It then seeks to justify a conclusion about the second case on the basis of the considerations about the first. (Note that not all usages of analogy contain argument.)

Analogue subject: In an argument by analogy, the **thing to which the primary subject is compared** and on the basis of which the arguer reasons to the conclusion about the primary subject. Some arguments by analogy use several analogues.
Primary subject: In an argument by analogy, the topic that the *conclusion* is about.
Consistency Analogy: Consistency analogies concern conclusions about how we should classify things (including states of affairs). They look for features of things (F, G, H) that **justify** a classification or decision or action (J). In the appeal to consistency, the analogue may be entirely hypothetical or fictitious without undermining the force of the argument.

P1: The analogue subject has features F, G, H, which merit the judgment J
P2 The primary subject has the same or similar features F', G, H'
C The primary subject merits the judgment J

Refutation by Logical Analogy – a subclass of consistency analogies that compares not things, but arguments.

P1: The analogue subject (an argument) has features F, G, H (such as structure), which merit judgment J (bad, not cogent, invalid)
P2: The primary subject (argument) has features F', G', H'
C: The primary subject (an argument) merits judgment J (bad, not cogent, invalid)

Inductive analogy: Inductive analogies concern conclusions about what is likely to occur (or has occurred) in the world. They look for features of things (F, G, H) that are likely to **cause** an event or state of affairs (Z). The analogue must usually be a real case (or real type of case), and the factual features of the analogue and the primary subject are essential for determining the strength of the argument.

P1 The analogue subject has features F, G, H which are causally relevant to it being Z
P2 The primary subject has same or similar features F', G', H'
C The primary subject is also Z

<u>Exercises</u>
For all the following exercises, identify any arguments by analogy, specifying the primary and analogue subjects. Determine what type of argument from analogy it is. Set out the argument with premises and conclusion. Is the argument cogent? Pay particular attention to relevant similarities and differences between the primary and analogue subjects.

(Exercise a) Govier chapter 11, pp 335-38, exercise 2, Q.6
Last week I sold my 1995 Toyota Tercel for $1900. The body was in good shape and so was the engine, and the mileage on this car is good, which is important given the rising cost of gas. Your 1995 Toyota Tercel is in good condition, just the way mine was, and the mileage will be the same. Therefore probably you can get around $1900 for your car, if you sell it soon.

(Exercise b) "Studying French without ever speaking the language does not result in a good command of French. Latin is a language too, so studying Latin without ever speaking it will not result in a good command of Latin."

(Exercise c) Govier, chapter 11, pp345-349. Excerpt from Exercise 3, Q.12
[The following passage is taken from a letter by Chris Brads to the editor of Globe and Mail (May 24, 2003) concerning legal penalties for possession of marijuana.]

"How can any public prosecutor stand before a judge with a straight face and argue that, while possession of 15 grams of pot in a plastic bag is not a criminal offence, owning the same amount in plant form is? If simple possession for personal use is not a criminal act, common sense dictates that growing marijuana for personal use should be treated the same way."

(Exercise d) We don't accept that Pepsi have the best brand of cola just because a celebrity such as Jennifer Lopez (J-Lo) says so. Similarly, why should we accept that the fault with our ASUS laptop is a burnt out hard-drive just because the ASUS technician said so?

(Exercise e) Govier Ch. 11, pp. 335–38: Exercise 2, Q. 5
In Plato's imaginary story of the cave, prisoners were in a cave where, in fixed positions, they saw only shadows and images in a darkened environment. They were unable to turn around and see the real world in the light of the sun. Plato's point was that people who saw only images would never know reality. We can apply the same sort of point to television shows today. The real world cannot be known from the experience of a screen from a stationary position on a couch.

(Exercise f) In 1986, a Canadian cabinet minister, John Crosbie, said that people in Canada's Atlantic provinces should not complain about the high level of unemployment in their region. After all, he said, they were well-off compared to people in Third World countries. Crosbie said that people in these areas should compare themselves, not with Canadians in central and western Canada, but rather with citizens in poorer parts of Asia and Africa. Calgary Herald communist Alan Connery satirized Crosbie's comments by saying that it is as if someone defended the performance of then Prime Minister Brian Mulroney, by pointing out that he was better than Hitler and Idi Amin.
[Amin was a former Ugandan ruler responsible for thousands of tortures and murders. You are to assess Crosbie's response.]

(Exercise g) "Slavery was a human institution that existed for thousands of years. Now slavery is largely eliminated. Morally dedicated citizens in religious and public action groups brought about the end of slavery, even though it was taken for granted as a natural fact of human existence. War, similarly, is a human institution that is taken for granted and that has existed for thousands of years. Morally dedicated citizens in a wide variety of groups, including many that have a religious basis, are working for the elimination of war. And the example of slavery shows that they can succeed."

(Exercise h) Govier, Ch. 11, pp. 329–33: Exercise 1, Part B: Q. 6
Background: The seventeenth-century philosopher René Descartes advocated a method of doubt. To build up a system of knowledge, he said, one should begin by doubting all his or her previous beliefs. Some critics objected to this method of doubt, saying that it was unrealistic and extreme. Defending it, Descartes said the following:

> "Suppose that a man had a basket of apples, and fearing that some of them were rotten, wanted to take those out lest they might make the rest go bad, how could he do that? Would he not first turn the whole of the apples out of the basket, and look them over one by one, and then having selected those which he saw not to be rotten, place them again in the basket and leave out all the others?"

<u>Week 11: Inductive Arguments: Sampling, Correlations, Causal Arguments, Analogies</u>

(Exercise a) Govier, Ch. 9, pp. 265-68: Exercise 1: Part A: Q. 2; Part C: Q. 2

(Exercise b) Comment on the reasoning in the following passage, identifying any fallacies.
"The Australian Bureau of Statistics reports that those with philosophy qualifications are three times more likely to bike to work than those who don't have philosophical qualifications. I am not surprised to learn this. The two people I know with philosophy degrees both cycle to work: they claim that it gives them room to think. It is clear that the mindset brought about by doing a philosophy degree makes you think outside the car-box: you want to get on that bike and pedal."

(Exercise c) Govier Ch. 10, pp. 303–4: Ex. 2: Q. 4.

(Exercise d) Govier, Ch.10, p 311, Ex.3: Q.5

(Exercise e) For the following argument, indicate whether it is inductive. If it is, then say whether it is an inductive analogy or an inductive generalisation or a causal induction, or some other inductive argument. Briefly evaluate the argument.
Chimpanzees and other higher primates have techniques of reconciliation after violent episodes and are able to manage conflict so that they can continue to live in reasonably harmonious social groups.

They have an instinctual base that provides for this conciliatory behaviour. Human beings have a common evolutionary history with these primates. Therefore they too have an instinctual base that provides for conflict resolution after quarrels. [Adapted from Frans de Waal, Peacemaking among Primates]

Extra from part (to supplement Exercise a):
Govier, Ch. 9, pp. 265: Exercise 1: Part C: Q. 3

Week 12: Conductive Arguments: Ethical and Inductive

There is no communal workshop on this topic. You are expected to work through the following questions independently and then check the answers when they come up on LMS. Attached is a copy of an article headed "The Give and Take Game". It is an article by Anjana Ahurja which first appeared in the Times newspaper. For convenience, its paragraphs have been numbered from 1 to 10. Read the article, and the Appendix, which contains Supplementary Notes, before the workshop, and answer the questions that follow the article.

Note: [for the very keen]: the article refers to a paper by three scientists: 'The Economics of Fair Play', by K. Sigmund, E. Fehr and M.Nowark, in *Scientific American*, N.Y., Jan. 2002.

1. *Paragraph 1*. Is the human behaviour, in the festive season, as described, a clear example of unselfish behaviour?

2. Paragraph 2

(i) It is stated that Darwinian Evolution rules that 'selfish is good", and that people are programmed to behave in ways that benefit them and their genes.

Are there problems with this claim? To help, think about these questions:

(a) what is it to be selfish? to act in ways that are in fact in your self-interest, or to act in ways which you believe to be in your self-interest?

(b) what is it for people to be "programmed to behave in certain ways"?

(ii) It is said that economists have "devised a description of this species _ Homo economicus _ an individual motivated by the need to maximise personal gain".

Are there any problems with this claim.

(iii) The third statement refers to certain "predictions" and to "irrational behaviour".

What are the predictions?

Why is the behaviour supposed to be irrational?

3. Compare paragraph 2 with paragraph 2* (taken from the more detailed article by in *Scientific American*) What important differences are there between the two paragraphs?

4. *Paragraph* 5 reports an experiment which is based on the discussion in *Paragraphs* 3 & 4. There is an important ambiguity in what "the experiment" is. What is the ambiguity? Why is it important to clarify this ambiguity? [There may be more than one ambiguity.]

5. *Paragraphs* 6 and 8 give a possible answer to a certain question/problem, one based on "central tenets of social interaction".

(a) What is the question or problem to which this is the answer?

(b) How is evolution supposed to be relevant to this answer?

(c) Are there difficulties with the answer?

(d) Are there other possible solutions?

6. What is the connection between the claim made in the final paragraph (10) to the rest of the article? For example, is the claim supported by the experiments described? (or does it have a different role?)

7. Look again at the sub-heading of the article. In the light of your reading of the article, is there anything wrong with the sub-heading?

The Give and Take Game

Are we generous by nature or genetically wired for selfishness? Researchers have devised a game to find out, says Anjana Ahurja [The Times]

1. The festive season is the time of giving and human beings are extraordinarily good at it. When we open Christmas presents from friends, we rarely wish their offerings to be much more expensive than ours, even if it means us "winning" in the seasonal transaction. In fact, our species appears to be so unselfish that scientists have been struggling to explain this human trait. But researchers in Austria, Switzerland and the US think they have come up with an answer.

2. Darwinian evolution rules that selfish is good and people are programmed to behave in ways that benefit them and their genes. Economists even devised a description of this species — Homo economicus — an individual motivated by the need to maximise personal gain. However, an intriguing game invented by economists and tested on various communities around the world contradicts such predictions and has become a prime example of irrational behaviour.

3.1 The Ultimatum Game goes like this. A stranger offers you $100. There are only two conditions attached to the offer. The first is, you have to share the cash with a second, anonymous person, say Mr B, who is sitting in another room and who you need never meet. You can give him anything between nothing and $100. The second condition is that Mr B, who knows about the $100, can decide either to accept or reject your offer. If he rejects it, both of you walk away with nothing. This is a one-play game: the first offer is final and there is no negotiation.

3.2 How do you play it? You could offer Mr B a small amount, say $10. It's a paltry sum, but he might think it's better than nothing. Or, he might become annoyed at you pocketing $90 and reject the deal, meaning that you both lose out. You could split it half and half, reasoning that this offers the fairest result. You could even offer more than $50, which would almost certainly secure the cash for both of you.

4.1 A poll of five colleagues revealed a very divided office — three would split the money equally "because it's fair", while two would offer $10 or less on the basis that only a "spiteful" or "illogical" person would scupper the deal. However, the latter two would revise their miserly policies if more was at stake, such as $10,000..

One would offer $5000, while another, around $3000.

4.2 In a more rigorous survey, Karl Sigmund a mathematics professor at the University of Vienna, Ernest Fehr, University of Zurich, and Martin Nowak, Institute of Advanced Study, Princeton, discovered that two-thirds of people placed in this position would give Mr B between $40 and $50. Only 4 per cent offered less than $20, and the vast majority of such low offers was rejected. The findings were irrespective of age, sex and even income.

5. The trio of academics, who discuss their experiment in Scientific American next month, find the rejections baffling.

> "Why should anyone reject an offer as 'too small'?" they write. "Mr B has just two choices: take what is offered or receive nothing. The only rational option for a selfish individual is to accept any offer. Even $1 is better than nothing ... But this is not how most people play the game ... Why do we place such a high value on fairness that we reject 20 per cent of a large sum solely because the co-player gets away with four times as much?"

The results, intriguingly, vary among communities. In the Machiguenga Amazonian tribe, for instance, the average offer is $26 rather than $45. Conversely, members of the Au tribe in Papua New Guinea routinely want to give away more than half of the money. In both cases, however, low offers were rejected.

6. The scientists have come up with a possible answer based on central tenets of social interaction — secrets, gossip and reputation. Our impulse to offer, and expect, a fair share of the pie has been sharply honed through evolution. Why? If word gets out that Mr B has accepted a low share, other people outside may surmise he is a soft touch. So, rather than risk this, Mr B may reject low offers for

the sake of his reputation. He may walk away with nothing but he has preserved his self-esteem and increased the chance of securing better offers in the long term.

7.1 Nowak, an evolutionary biologist, designed a computer program to test the idea from an evolutionary angle. He created a population of simulated people who were let loose to play the Ultimatum Game with each other. The players who won the most money were programmed to reproduce faster than those who gained less. Sure enough, those players who offer the lowest sums ended up dominating the field.

7.2 However, when the program was tweaked so players could get information about the past behaviour of other players, the scenario changed. Mean players, who offered very small amounts, no longer prospered. Instead, players who offered fair sums and rejected paltry ones came to dominate. A reputation for playing fair then offered an evolutionary advantage.

8. Whether this reasoning [in paragraph 6] goes through a human being's head is debatable – many people would have a strong, instinctive emotional response to being "diddled". This emotion, argue the researchers, has been sharpened by evolution over hundreds of thousands of years, allowing us to automatically override an acceptance of anything less than a fair split without necessarily being aware of the machinations behind it. It is also likely that the guilt or shame associated with offering low sums has specifically evolved to stop us from offering parsimonious settlements.

9. The researchers say that economists were wrong to "study their selfish Homo economicus" and wrong to imagine that each of us is a Robinson Crusoe on a desert island, unmoved by social interactions. "We are no Robinsons," they conclude. "Our ancestors' line has been social for 30 million years. And in social interactions, our preferences often turn out to be far from selfish."

10. "Ethical standards and moral systems differ from culture to culture, but we may presume they are based on universal, biologically-rooted capabilities, in the same way that thousands of different languages are based on a universal language instinct."
The Weekend Australian, "Inquirer", January 5-6, 2002, p.20.

APPENDIX: Supplementary Notes

1. Compare *Paragraph 2* with *Paragraph 2** (taken from the article 'The Economics of Fair Play', by K. Sigmund, E. Fehr and M.Nowark, in *Scientific American*, N.Y., Jan. 2002.)

> 2*. "For a long time, theoretical economists postulated a being called Homo economicus – a rational individual relentlessly bent on maximising a purely selfish reward. But the lesson from the Ultimatum Game and similar experiments is that real people are a cross-breed of Homo economicus and H.emoticus, a complicated hybrid species that can be ruled as much by emotion as by cold logic and selfishness. An interesting challenge is to understand how Darwinian evolution would produce creatures instilled with emotions and behaviours that do not immediately seem geared toward reaping the greatest benefit for individuals or their genes."

2. Evolutionary psychology
Paragraph 8 presents an evolutionary explanation. This explanation draws upon the field known as *Evolutionary Psychology*. To help in this exercise, let us assume that Evolutionary Psychology implies the following:

> "One's genetic endowment provides us with the propensity to develop psychological states (emotions, feelings, desires, . . .) that dispose us to act in ways that serve to increase the chances of one's genes surviving/flourishing – at least in the types of circumstances that existed in the beginnings of human social life."

<u>Also in Week 12</u>: Be sure you have marked the second test.